Fiscal Year:	FY 2014	**Task Last Updated:**	FY 07/23/2014
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	10/01/2009	**End Date:**	12/11/2014
No. of Post Docs:	3	**No. of PhD Degrees:**	6
No. of PhD Candidates:	1	**No. of Master' Degrees:**	5
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 12/11/2014 per award of new continuation grant, per NSSC (Ed., 9/30/2015) NOTE: Project extended to 9/30/2015, per D. Stocker/GRC and NSSC, pending award of new grant (Ed., 10/6/14) NOTE (Ed., 7/23/2014): This project is being continued as a new grant to be awarded in fall 2014. See that task for subsequent reporting. NOTE: Project extended to 9/30/2014 (NSSC information/Ed., 7/2013) NOTE: Project to be extended to 9/30/2014 per D. Stocker/GRC (Ed., 7/20/2011)		
Key Personnel Changes/Previous PI:	Not applicable.		
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii at Manoa) Sunderland, Peter (University of Maryland) Urban, David (NASA Glenn Research Center)		
Grant/Contract No.:	NNX09AO53A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI, the three Co-Is, and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, Peter Sunderland directing numerical simulations at University of Maryland, and Beei-Huan Chao directing analytical work and computations at University of Hawaii.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	Summary of Accomplishments: • The research team published 4 peer-reviewed journal articles during this period. This work is summarized below. • Work at Washington University continued to pursue normal gravity experiments of soot formation in diffusion flames, numerical modeling, and chemical kinetics. • Work at Univ. of Maryland continued to pursue numerical models of spherical diffusion flames, including soot formation and extinction. • Work at Univ. of Hawaii continued to pursue analytical and computational studies of soot formation, radiation, and extinction in diffusion flames. • The flight experiment was supported in various ways. For example, the research team responded to issues raised by NASA engineers in refining the test matrix and ACME requirements document. The research team also aided staff at NASA Glenn with burner characterization drop testing in the 5 s drop facility. V.R. Lecoustre, P.B. Sunderland, B.H. Chao, R.L. Axelbaum, Modeled Quenching Limits of Spherical Hydrogen Diffusion Flames, Proceedings of the Combustion Institute 34 (2013) 887-894. Hydrogen-air diffusion flames were modeled with an emphasis on kinetic extinction. The flames were one-dimensional spherical laminar diffusion flames supported by adiabatic porous burners of various diameters. Behavior of normal (H2 flowing into quiescent air) and inverse (air flowing into quiescent H2) configurations were considered using detailed H2/O2 chemistry and transport properties with updated light component diffusivities. For the same heat release rate, inverse flames were found to be smaller and 290 K hotter than normal flames. The weakest normal flame that could be achieved before quenching has an overall heat release rate of 0.25 W, compared to 1.4 W for the weakest inverse flame. There is extensive leakage of the ambient reactant for both normal and inverse flames near extinction, which results in a premixed flame regime for diffusion flames except for the smallest burners with radii on the order of 1 micron. At high flow rates H+OH(+M)?H2O(+M) contributes over 40% of the net heat release. However at flow rates approaching quenching limits, H+O2(+M)?HO2(+M) is the elementary reaction with the largest heat release rate. The corresponding author for this paper was P.B. Sunderland. F. Xia, R.L. Axelbaum, Simplifying the complexity of diffusion flames through interpretation in C/O ratio space, Computers and Mathematics with Applications, 65 (2013) 1625-1632. Understanding the structure of diffusion flames is often complicated by the dependence of flame structure on the boundary conditions, e.g. composition, temperature and flow field (e.g., strain rate in a counterflow flame.) In this paper, the local carbon-to-oxygen atom ratio (C/O ratio) is applied as a variable to interpret the flame and soot zone structures of counterflow diffusion flames from numerical results with detailed chemical kinetics and transport. Radical pool and soot precursor zones are shown to be clearly delineated in C/O ratio space. The boundary of these two zones, as well as the flame location, are shown to be independent of both stoichiometric mixture fraction (Zst) and strain rate when interpreted in C/O space. The kinetic ratio is used to study the characteristics of key chemical reactions and to identify regions of equilibrium for these reactions. The results of this paper indicate that the C/O ratio is a valuable variable for interpreting flame structure and soot precursor chemistry for diffusion flames. The corresponding author for this paper was R.L. Axelbaum. F. Xia, G.S. Yablonsky, R.L. Axelbaum, Numerical Study of Flame Structure and Soot Inception Interpreted in Carbon-to-Oxygen Atom Ratio Space, Proceedings of the Combustion Institute, 34 (2013) 1085-1091. The utility of interpreting flame results in carbon-to-oxygen atom ratio (C/O ratio) space, as opposed to physical space or mixture fraction space, is evaluated. Flame and soot zone structures of counterflow diffusion flames have been studied for C2H4 and C3H8 and interpreted in C/O ratio space as a function of stoichiometric mixture fraction (Zst). The Burke-Schumann results expressed in C/O ratio space demonstrate how a clear and direct understanding of how structure is affected by Zst can be realized in C/O ratio space because, unlike physical or mixture fraction space, the flame location is independent of stoichiometric mixture fraction. Numerical results with detailed chemical kinetics also indicate that C/O ratio space is a fundamental variable in the sense that, for a given fuel, the location of the flame zones and critical reactions is invariant with Zst and strain rate. Two zones are clearly observed, namely the radical pool and the soot precursor zone located on the fuel side of the flame. The onset threshold of soot precursors (C6H5 and C6H6) for the high temperature side of the soot precursor zone is characterized by the depletion of radicals. The role of the hydrogen radical in flame structure and soot inception is demonstrated by studying its production and consumption channels in C/O ratio space. Finally, a modified (C/O)* is given to interpret the physical meaning of C/O ratio. The numerical results in this work indicate and explain the advantages of applying C/O ratio space in the analysis of flame structure and soot precursor chemistry. The corresponding author for this paper was R.L. Axelbaum. F. Yi, R.L. Axelbaum, Stability of Spray Combustion for Water/Alcohols Mixtures in Oxygen-Enriched Air, Proceedings of the Combustion Institute, 34 (2013) 1697-1704. The feasibility of burning fuels that are heavily diluted with water was evaluated. This concept could potentially be utilized to directly generate steam for a variety of applications, including power generation and enhanced oil recovery. Two water-soluble fuels, ethanol and 1-propanol,

	were heavily diluted with water and sprayed into a swirl-stabilized burner and the blow-off limits were obtained to identify the flame stability. The blow-off limit was experimentally determined as the lowest O2 concentration where a flame could exist under a given oxidizer flow rate. For the present experimental system, stability maps were determined as a function of fuel concentration and oxidizer flow rate for different types of nozzles and both alcohols. Contours of temperature and overall fuel mass fraction in the droplets were obtained for a flame burning 15 wt% ethanol in an oxidizer of 50% O2 and 50% N2. The results revealed that preferential vaporization of ethanol over water plays an important role in enhancing flame stability. To further understand this, a model of bi-component droplet vaporization was employed for both the diffusion limit and distillation limit modes of liquid mass transfer. The results confirmed the preferential vaporization of ethanol and demonstrated that it is necessary to account for the non-ideal solution behavior of the alcohol-water mixture to appropriately model the vaporization process. The model also demonstrated that for this experimental system, droplet vaporization was better characterized by the distillation limit mode with enhanced mass transfer by convection. The corresponding author for this paper was R.L. Axelbaum. NOTE (Ed., 7/23/2014): This project is being continued as a new grant to be awarded in fall 2014. See that task for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Abstracts for Journals and Proceedings	Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Effects of Temperature and Equivalence Ratio on Soot Precursors in Rich Premixed Flames." Poster. Eighth U.S. National Combustion Meeting, Salt Lake City, Utah, May 2013. Eighth U.S. National Combustion Meeting, Salt Lake City, Utah, May 2013. , May-2013
Articles in Peer-reviewed Journals	Lecoustre VR, Chao BH, Sunderland PB, Axelbaum RL. "Extremely weak hydrogen flames." Combustion and Flame. 2010 Nov;157(11):2209-10. http://dx.doi.org/10.1016/j.combustflame.2010.07.024 , Nov-2010
Articles in Peer-reviewed Journals	Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Modeled quenching limits of spherical hydrogen diffusion flames." Proceedings of the Combustion Institute. 2013;34(1):887-94. http://dx.doi.org/10.1016/j.proci.2012.07.029 , Jan-2013
Articles in Peer-reviewed Journals	Xia F, Axelbaum RL. "Simplifying the complexity of diffusion flames through interpretation in C/O ratio space." Computers and Mathematics with Applications. 2013 May;65(10):1625-32. http://dx.doi.org/10.1016/j.camwa.2013.01.008 , May-2013
Articles in Peer-reviewed Journals	Xia F, Yablonsky GS, Axelbaum RL. "Numerical study of flame structure and soot inception interpreted in carbon-to-oxygen atom ratio space." Proceedings of the Combustion Institute. 2013;34(1):1085-91. http://dx.doi.org/10.1016/j.proci.2012.06.043 , Jan-2013
Articles in Peer-reviewed Journals	Yi F, Axelbaum RL. "Stability of spray combustion for water/alcohols mixtures in oxygen-enriched air." Proceedings of the Combustion Institute. 2013;34(1):1697-704. http://dx.doi.org/10.1016/j.proci.2012.05.088 , Jan-2013